METOCLOPRAMIDE

Extrapyramidal effects

Max 5 days

CHEMICAL NAME 4-amino-5-chloro-N-[2-(diethylamino)ethyl]-2-methoxybenzamide (hydrochloride)

CLASS Dopamine D2 antagonist antiemetic and prokinetic

TRADE NAMES Maxolon · Reglan · Primperan · Plasil

THERAPEUTIC USES / INDICATIONS

- Nausea and vomiting (post-operative, chemotherapy-induced, migraine-associated, radiotherapy).
- Gastroparesis (diabetic) — short-term.
- Facilitates small-bowel intubation and radiology.
- Migraine (with analgesic, improves absorption).

ADMINISTRATION

- **IV** slowly over at least 3 min (rapid → anxiety, restlessness, hypotension).
- **PO** 30 min before meals; **IM**.
- Limit use to 5 days (tardive dyskinesia risk).

DOSAGES

- **Adult:** 10 mg up to three times daily (max 30 mg/day or 0.5 mg/kg/day).
- **Children (1–18 yr, 2nd-line only):** 0.1–0.15 mg/kg up to three times daily.
- **Renal impairment (CrCl <15):** 75% reduction; moderate: 50%.
- **Elderly:** lower dose.

PHARMACOKINETICS

- **Onset:** IV 1–3 min; PO 30–60 min. **Duration:** 1–2 h (antiemetic longer).
- Metabolised in liver; crosses blood–brain barrier.
- Excreted in urine. **t½:** 5–6 h.

PHARMACODYNAMICS

- Blocks **D2 receptors** in CTZ → antiemetic.
- Peripheral D2 block + 5-HT4 agonism → ↑ gastric emptying and lower oesophageal sphincter tone (prokinetic).
- At high doses: 5-HT3 blockade.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Drowsiness, restlessness, fatigue.
- Diarrhoea.
- Raised prolactin (galactorrhoea, menstrual changes).

ADVERSE EFFECTS (SERIOUS)

- Acute dystonic reactions (oculogyric crisis, torticollis); young women and children; treat with procyclidine or diphenhydramine.
- Akathisia; tardive dyskinesia (long-term, often irreversible).
- Neuroleptic malignant syndrome.
- Arrhythmias with rapid IV; depression.

CONTRAINDICATIONS & CAUTIONS

- GI obstruction, perforation or haemorrhage; within 3–4 days of GI surgery.
- Phaeochromocytoma; epilepsy.
- Parkinson's disease; history of tardive dyskinesia.
- Children <1 yr; use with levodopa.

KEY INTERACTIONS

- **Levodopa, dopamine agonists:** mutual antagonism.
- **Antipsychotics:** ↑ EPS.
- **Opioids, anticholinergics:** antagonise prokinetic effect.
- **SSRIs:** serotonin syndrome (rare).

NURSING MANAGEMENT

- Give IV slowly over ≥3 min.
- Monitor for dystonia, restlessness, abnormal movements; antidote available.
- Check for bowel obstruction before giving.
- Limit duration; avoid in Parkinson's.

PATIENT EDUCATION

- Report muscle spasms, eyes rolling upward, stiff neck, restlessness.
- Do not use longer than prescribed.
- May cause drowsiness.

KEY POINTS

- D2 antagonist — antiemetic + prokinetic.
- Dystonia (young), tardive dyskinesia.
- 10 mg TDS, max 5 days.
- Never in bowel obstruction or Parkinson's.

SOURCES CHECKED **[530]** Metoclopramide 5 mg/5 ml Oral Solution Sugar-Free (UK SmPC) **[531]** Metoclopramide 5 mg/ml Injection (UK SmPC) **[532]** Metoclopramide, IM, IV (US label)

Full references p. 432–457